

Colonizing Pirkanmaa – The Game

Game description and rules:

Colonizing Pirkanmaa is a 2-4 multiplayer boardgame. Your objective is to be the only ruler of Pirkanmaa. You must use your units to conquer tiles in the map. You can also construct different buildings and harvest resources to gain advantage to your enemy. After you have settled down you can start conquering enemy tiles. Capturing 70% of the whole area or conquering the enemy headquarters is the key to the victory. Remember to manage your resources correctly or you will lose the game.

Map consists of tiles. Size of map can be chosen at the start of the game. Only one building can be built to each tile (if tile allows building). There can be only 3 units in each tile. One exception is when tile is getting conquered when there can be a total of 6 units in one tile. Each players region is surrounded with lines corresponding to their colour.

Main game mechanics are wrapped around managing resources, managing unit counts and having good conquering strategies. A good player can and will predict enemy movements and strategies far before they happen.

Winning:

You will win the game if you are the last man standing.

Losing:

You will lose when: your resources are not enough to pay your expenses; your headquarters gets conquered by enemy soldier or someone else owns 70% of all tiles.

Start of the game:

At the start of the game players can choose the starting tile. Starting tile must be a grassland. Player will also get the ownership of tiles around the chosen tile if they are still unowned. Players headquarters is placed in the chosen tile.

Resources:



Money: Key resource for buying buildings and units.



Wood: Can be harvested from the forest. Wood is used to build pretty much everything.



Stone: Can be harvested from mines. Used for buildings too.



Metal: Needed for the advanced stuff. You can get metal from mines too.

Tiles:



Grassland:

Grassland is the basic tile of the game. You can build many buildings to this tile. Available buildings are: Farm, Village, Outpost and Nuclear Power Plant. Grassland has no additional effects.



Forest:

Forest is the only way to get some wood. You will also get some stone when forest is cut down. Forest can be harvested for 6 times before it is fully cut down. When forest is fully cut down you can build on top of the tile (same buildings as grassland). It takes 5 turns to forest to grow back. If many workers are cutting down forest it will cut down faster.



River:

River can be problematic tile sometimes. You can move your units to this tile, but you cannot move further before building a bridge or hydroelectric power plant.



Mountain:

Mountain is the perfect place to build the mine. That is the only building that can be build to mountain. Mountains are super important.



Picture 1: The game in action



Abundant Forest:

Abundant Forest can be harvested for fruits and other goods. Only one worker is needed to do so. This is the only tile where you can make some money without building anything.

Units:

Units can be bought from the unit shop. You can only buy units up to your maximum unit limit. Your maximum unit limits can be seen in the menu. You can get bigger maximum unit limit by building or conquering buildings that grant more maximum unit limit. There is different limit for soldiers and other units.



Worker:

Worker is the basic unit for everything. It can be used to work in the buildings, and it can harvest forests. Workers are the least expensive units available. You must pay a little salary though.



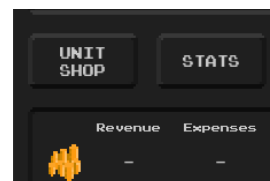
Expert:

Expert is required for some more advanced work like for power plants. It can also boost workers to work more efficient. Experts are expensive to buy and require more salary.



Soldier:

Soldier is for protecting your tiles and conquering other players tiles. Soldiers are expensive and require some metal too. Their salary is very high too.



Picture 2: Unit shop can be opened in the menu.



Picture 3: You can see your units and maximum unit limits in the menu.

Buildings:



Headquarters:

Headquarters is the heart of your region. Only one headquarters can be built in the start of the game. Building headquarters will give player the ownership of all the other tiles around the headquarters too. If enemy manages to place a soldier to your headquarters, you will lose. Building headquarters is free. Headquarters also grant:

- +3 Max Units
- +1 Max Soldier



Village:

Village is the way to increase your maximum units. Remember to protect this since if it gets conquered your units might lose homes and even get eliminated too. Can be build on grassland and cut forest.

- +3 Max Units



Farm:

Farm is good way to get some money. Crops can be harvested every 4 rounds granting decent amount of money. If worker leaves farm in the middle of growth, crops will die, and everything has to be started all over again. Can be build on grassland and cut forest. Adding more workers will not increase the revenue coming from the farm.



Mine:

Mine is a super important building since it is the only way to get some consistent stone and metal. Building mine also grants player some metal and stone. Many workers can work in mine at the same time each multiplying the revenue. Expert working in mine will double whole the revenue. Some money and wood are required to build a mine. Mine can only be built in the mountain.



Mikontalo:

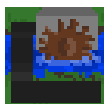
There is only one Mikontalo in the game. It will spawn randomly in the start of the game. Conquering Mikontalo will grant player:

- +2 Max Units



Nuclear Power Plant:

This building is the ultimate money maker. Gives consistent money every round. One expert and worker are required to run this thing. Every worker will multiply the revenue. This can only be built on grassland and cut forest.



Hydroelectric Power Plant:

Second best way to make some money. Gives consistent money every round. Also acts as a bridge. Can only be build to straight river tile. Requires one expert and worker just like the Nuclear Power Plant. Every worker will also multiply the revenue.



Bridge:

Bridge can be built to straight river tiles. Bridge allows player to move further from the river tile. Building a bridge requires some money, wood and stone. Some wood is required every round to maintain bridge in the good condition.



Outpost:

Outpost is a perfect building for defending your region, since outposts cannot be conquered. Friendly outposts cannot be built directly next to each other or next to your headquarters. Can be build on grassland and cut forest. Outpost also grants player:

- + 3 Max Soldiers

Conquering mechanics and policy:

Main way to overthrow your enemy is by conquering your enemy's tiles. All the conquering units are put to upper part of the tile like in picture 4.

Every type of unit can be used to conquer tiles in the map. Tiles can only be conquered one tile away from your current region. Tile can be conquered with any unit if tile is not owned by anyone. However, if tile is owned by another player soldier is required to conquer the tile. if you have more soldiers in the tile you will win. Otherwise you will lose and your conquering units will get eliminated. Remember that enemy outposts cannot be conquered.

Regions can also be cut. If you manage to conquer tiles in a way that enemy region is disconnected from the headquarters, tiles that were disconnected are now not owned by anyone. All the units in the disconnected area will get eliminated but buildings can be reused.



Picture 4: Conquering units are in the upper part of the tile.



Picture 6: Red players area before cut by Blue player



Picture 7: The mine remains but Reds unit got eliminated.



Picture 5: Units can be moved only one tile away from your region.

Conquering other players headquarter will fully eliminate the other player. All the other players tiles are now owned by the conqueror. Other players units are eliminated, and resources are discarded but all the buildings will remain in the tiles and can be reused by the conqueror.



Picture 8: Red is about to conquer Blues HQ.



Picture 9: Blue got eliminated and Red gets Blues tiles.

Menu and game controls:

Starting screen:

Starting menu is the very first window that will open when game is launched. There are some game options that can be determined. First there are map width and height (in tiles). Map seed can also be chosen for different maps. There are 15000 different maps in total. You can choose how many players will play the game and even type names for the players.

10	Map width (between 10 and 25)
10	Map height (between 10 and 15)
139	Map seed (between 1 and 200)
<button>Randomize map seed</button>	
2	Players (between 2 and 4)
PlayerOne	PlayerTwo
PlayerThree	PlayerFour
Enter player names in the boxes above	
Start	Exit

Picture 10: The starting menu

Default menu:

The first and the most important menu is the default menu. It contains a lot of information related to the current player. It shows current players name and the corresponding colour in the top. Under the name there is resources which player currently has.

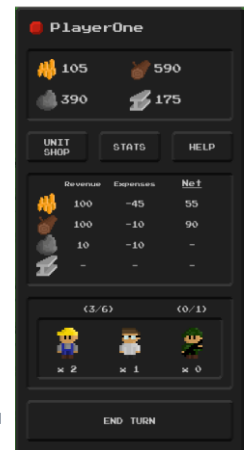
Under resources there are three buttons. First button will lead player to the unit shop. Second button will open the stats menu. And the last button will open game rules and help.

Next element is the net value chart. It will tell player in real time how much resources are spent and received after current round. This is very important since player can use this chart to check if there are enough resources to do some certain task.

Under the net value chart there is the unit view. Unit view will tell player how many units are in use and what are the maximum unit amount limits.

The last element is the end turn button which is pressed when the player has done everything he wanted for the current round.

Picture 11: Default menu



Tile inspection menu:

Tile inspection menu is the second most important menu in the game. When the tile is clicked it will open the tile inspection menu for the clicked tile. In the very top there is current players name and corresponding colour. Under that there is the tile description box. It will show the type of the tile or the building and a little description.

There is also an animated picture of the tile or the building. Under the picture there is a corresponding colour of the owner of the tile. If tile is not owner by anyone the colour is grey. Under the little description there is a current net value of the tile which indicates how much resources are gained or lost in the inspected tile. There is also some extra information about the tile but those are tile specific. For example, in the picture 10 there is a growth indicator for the farm.

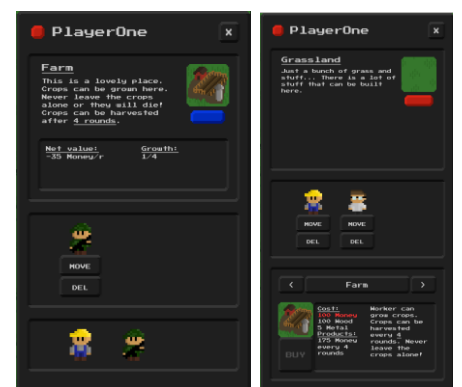
If there is ally units in the tile there is going to be the ally unit view under the tile description. Ally unit view will indicate which units are currently in the tile and there are buttons for moving or deleting the unit.

if there are enemy units in the tile there is going to be enemy unit view. This view has no functionality, but it will indicate if the inspected tile has enemy units in it.

Next element in the tile inspection menu is the build menu. Build menu is shown when there are buildings available to be build in the tile. If there are many buildings which can be built there are buttons to scroll different buildings left and right. If there are not enough resources to build certain.

building the text for the corresponding lacking resource is going to appear red. Buy button also changes colour if building cannot be built. There is a little description about the building too.

You can close tile inspection menu by clicking the inspected tile again or the close button in the top right corner.



Pictures 12 and 13: Tile inspection menus



Picture 14: Inspected tile will have special appearance in the map

Unit shop:

Unit shop can be opened by pressing the unit shop button in default menu. Again, there is current players name and corresponding colour in the very top. Under that there are the current resources player has. Then there is the shop view. There are all three different units available to buy. The text colour in the buy button will indicate if you can currently buy the unit. Next to the unit picture and the buy button there is a little description that includes the cost and how much salary must be payed when unit is in use. In the top right corner, there is the close button which returns player to default menu.

Stats menu:

Stats menu can be opened by pressing stats button in the default menu. Again, there is current players name and corresponding colour in the very top. After that there is how many rounds has been played so far. One round means that every player has made a move.

Under the rounds played section there is the tiles owned box. It will indicate how many percent of the tiles is owned by each player and how many tiles are not owned by anyone. This is a good tool for checking if any player is near getting 70% of all tiles and winning. You can return to the default menu by pressing the button in the top right corner.

Help / rules window:

Pressing the help button in the default menu will open help and rules dialog.

All the extra features in this project:

- Starting tile choosing mechanics. Players headquarters is placed to the selected starting tile and player will also get the ownership of all the other tiles next to chosen tile.
- Custom world generator: There are algorithms for spawning forests and mountains naturally. River will also spawn on the map. With custom seed players can play different maps. There are 15000 maps in total.
- Starting menu: Players can choose their own names at the start of the game. Map size and seed can be chosen too. Map tile size will scale depending of the map height. Menu scales according to the map size.
- New game mechanic: Conquering. Player can conquer tiles like described earlier in this documentation. Check out “Conquering mechanics and policy”.
- Nice custom graphics with animations. Animation options can be individually set for each object in the game easily. Graphics can be assigned for objects and changed while game is running. Theme for the game graphics is inspired by the old 8-bit games.
- Fully custom in game menus with custom elements like buttons, containers and labels. Graphics for the menu are also animated. Menus are highly interactive with the game. There are multiple different views for the menu for every situation possible.
- Well implemented user interface and game controls with easy to use menus, logical indicators and cursor following objects.
- 5 different tile types and 9 different building types. There are also 3 different types of units.
- Custom building and tile mechanics for each tile and building type. Using different units in the tiles will affects the outcome.
 - o Forest can be cut and regrown. Buildings can be built on top when cut.
 - o River blocks user from moving from tile to next if bridge or hydroelectric power plant is not built.
 - o Farm needs to be grown for 4 rounds before harvesting. You cannot leave the farm or crops will die.
 - o Expert is needed for power plants to run. Expert will also boost up mining efficiency.
 - o Soldiers can be used to defend or conquer tiles.
 - o Buildings like village and mikontalo will increase players maximum unit amount.
 - o Outpost cannot be conquered and will increase the maximum soldier amount.
- In game live statistics to make the gaming experience better.
- 2 – 4 multiplayer game.

Main focus of the labour:

Otto Ranta-Ojala:

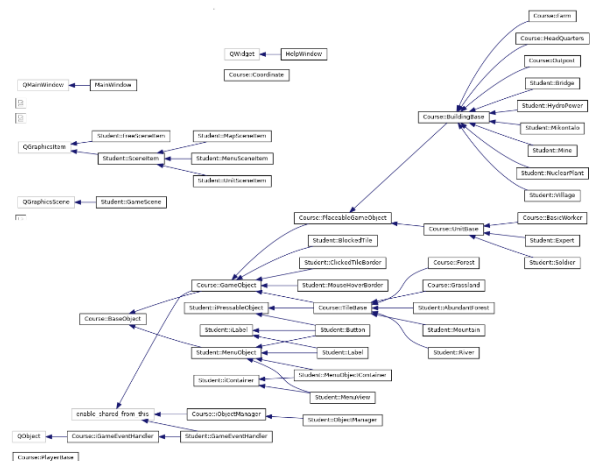
- Most of the basic game mechanics
 - o Implementation of the building rules
 - o Tile conquering mechanics (Fully implemented by Otto)
 - o Interactions between units, buildings and tiles
 - o Player resource management
- Players and player management
- Tile drawing and players region boundaries
- Starting window and help / rules window.
- Memory management and testing.

Tino Kaartovuori:

- Ideas for different types of buildings, tiles and units
- Designing most of the class hierarchy, DAL implementations and interfaces
- Hand drawn graphics and all the animations.
- All custom in game menus.
- Other in game overlays like cursor following objects.
- World generation
- Unit placement mechanics
- Resource and description maps
- Balancing the game

Checkout file: /Doxygen files/html/index.html

for more information about the class structure and relationships.



Picture 15: Class Hierarchy